```
!pip install gensim
import numpy as np
from gensim.models import KeyedVectors
from numpy.linalg import norm
import matplotlib.pyplot as plt
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in
/usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in
/usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in
/usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: wrapt in
/usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1-
>gensim) (2.1.0)
Downloading gensim-4.4.0-cp312-cp312-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 64.2 MB/s eta
0:00:00

Successfully installed gensim-4.4.0
```

```
print("Loading Word2Vec model...")
model = KeyedVectors.load_word2vec_format(
    "GoogleNews-vectors-negative300.bin",
    binary=True
)
print("Model loaded!")
```

```
Loading Word2Vec model...
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call
last)
/tmp/ipython-input-216292393.py in <cell line: 0>()
      1 print("Loading Word2Vec model...")
----> 2 model = KeyedVectors.load_word2vec_format(
      3     "GoogleNews-vectors-negative300.bin",
      4     binary=True
      5 )

/usr/local/lib/python3.12/dist-packages/gensim/models/keyedvectors.py
in load_word2vec_format(cls, fname, fvocab, binary, encoding,
unicode_errors, limit, datatype, no_header)
   1719
   1720         """
```

```
-> 1721         return _load_word2vec_format(
   1722             cls, fname, fvocab=fvocab, binary=binary,
encoding=encoding, unicode_errors=unicode_errors,
   1723             limit=limit, datatype=datatype,
no_header=no_header,

/usr/local/lib/python3.12/dist-packages/gensim/models/keyedvectors.py
in _load_word2vec_format(cls, fname, fvocab, binary, encoding,
unicode_errors, limit, datatype, no_header, binary_chunk_size)
   2059         else:
   2060             header = utils.to_unicode(fin.readline(),
encoding=encoding)
-> 2061             vocab_size, vector_size = [int(x) for x in
header.split()]  # throws for invalid file format
   2062         if limit:
   2063             vocab_size = min(vocab_size, limit)

ValueError: invalid literal for int() with base 10: 'PK\x03\x04'

# Target sets
X = ["man", "male", "boy", "brother", "him", "son"]
Y = ["woman", "female", "girl", "sister", "her", "daughter"]

# Attribute sets
A = ["career", "business", "office", "salary", "corporation",
"management"]
B = ["home", "family", "children", "parents", "wedding", "relatives"]
```